



# Chapter 5 Conditionals

CS 101- Programming  
Fundamentals

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# Chapter 5 Conditionals

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**New operators:** Floor division and modulus.

- `//` divides two numbers and rounds down to an integer.
- `%` divides two numbers and returns the remainder.

**Boolean expressions:** Either true or false

**Relational operators:** `==`, `!=`, `>`, `<`, `>=`, `<=`

**Logical operators:** `and`, `or`, `not`

# Chapter 5 Conditionals

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- **Conditional statements:** check conditions and change the behavior of the program accordingly.
- **If statement:**
  - **Compound statement:** header followed by intended body
  - **header:** *If* followed by *boolean condition*
  - **body:** *statement (s)*

```
if x < 0:  
    print ( x is negative)
```

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- **Alternative execution:** condition determines which of the two possibilities runs.
- **Branches:** if, else

```
if x % 2 == 0:  
    print ('x is even')  
else:  
    print ('x is odd')
```

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- **Chained conditionals:** express a computation requiring more than two possibilities /branches.
- **Branches:** if, elif, (else)

```
if x > 0:  
    print ('x is positive')  
elif x < 0:  
    print ('x is negative')  
else:  
    print ('x is zero')
```

```
if marks >= 95 and marks <= 100:  
    Grade = 'A'  
elif marks >= 90 and marks <=94:  
    Grade = 'A-'  
elif marks >= 60 and marks <=63:  
    Grade = 'C-'  
else:  
    Grade = 'F'  
print (Grade)
```

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- **Nested conditionals:**

- test a condition within another condition
- reduced readability

```
if x == y:  
    print('x and y are equal')  
else:  
    if x < y:  
        print('x is less than y')  
    else:  
        print('x is greater than y')
```